PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Quality of clinical practice guidelines for inadequate first-line treatment for depression according to AGREE checklist and comparison of recommendations: a systematic review
AUTHORS	Gabriel, Franciele; Stein, Airton; de Melo, Daniela; Fontes-Mota, Géssica C. H.; dos Santos, Itamires B.; de Oliveira, Aliandra F.; Fraguas, Renerio; Ribeiro, Eliane

VERSION 1 – REVIEW

REVIEWER	Bayes, A UNSW, Black Dog Institute
REVIEW RETURNED	19-Jul-2021

GENERAL COMMENTS	This paper seeks to examine CPGs for unipolar depression comparing and contrasting recommendations for patients with an inadequate response to first line treatment, using the AGREE-II tool. Overall while the paper was methodologically sound, the results found differences but were not particularly enlightening. Please justify why Recommendations for patients who did not respond to the first-line pharmacological treatment grouped according to patient reevaluation, dosing adjustment, switching, combination, and medication augmentation? Regarding method, it should specify if only unipolar depression is being examined (and bipolar depression is excluded) as there would be differences in recommendations between the two. The references in text seem to jump from 28 to 67. I could not see the references in superscript between these numbers. The Discussion makes reference to 'more clinical trials' needed. I am not sure how this would specifically assist in guidelines being more specific, please elaborate? The paper would benefit from discussion of limitations of guidelines themselves, that they are supposed to be 'evidence-based' but if the included evidence is of poor questionable quality then the guidelines will reflect this.

REVIEWER	Gerlach, Lauren University of Michigan
REVIEW RETURNED	02-Aug-2021

GENERAL COMMENTS	In this systematic review of clinical practice guidelines for patients who fail first line antidepressant treatment for depression, the authors highlight the significant variation and lack of consensus of current guidelines. Given the large impact of depression on quality of life for patients, costs of delivering care, and high prevalence of disease, as the authors highlight, more consensus is needed. Recommendations for improving the manuscript are below. Major Concerns: - My biggest recommendation would be to reformat the Discussion section to focus more on what are the most common recommended treatment approaches and components of treatment based on the available current clinical practice guidelines. Even given the large variability present, discussing in more detail the range of treatment recommendations I think would be a helpful take away for readers and help make it more clinically relevant. - Additionally, any revised Table or Figure that could better summarize this material would be helpful for the reader. As is, the Tables presented are so large that they become difficult to read across multiple pages Minor Comments: Abstract: - Page 2, line 7: Please adjust wording of the Objective sentence ("depression differ for adults"). Similarly, the Conclusions statement could be improved. - Page 3, line 6: Which two guidelines does this refer to? Introduction: - Page 4, line 22: For the authors statement: "Along with an increasingly aging population", I would challenge the authors that depression prevalence among community dwelling (non-institutionalized) older adults (65+ years) is actually lower than that of general adults. Additionally, studies out of the U.S. suggest that older adults are also less likely than general adults to experience depression, anxiety, and insomnia following the start of the pandemic. Your point that COVID has been associated with increased mental health symptoms globally and therefore highlights the need for treatment services is well taken. However, I would not limit this discussion to just older adults. Methods: - Page 5, line 55: It would be helpful for readers not familiar to clearly specify what primary and secondary care for depression is. Does this include visits to outpatient psychiatrists or just primary care physicians?

REVIEWER	Ball, Patrick University of Wolverhampton, School of Pharmacy
REVIEW RETURNED	11-Oct-2021

GENERAL COMMENTS	Thank you for the opportunity to review this interesting article. This review was undertaken in a competent and rigorous manner, and highlights interesting limitations in the guidelines currently available and publication would be helpful. Areas where I would suggest minor revision are:

	It is noted that there was high heterogeneity in some AGREE domains. Since many guidelines take up to five years from inception to publication, and DSM iV was replaced by DSM V in 2013, could some of this be due to variations in definitions applied? Perhaps this could be mentioned in the discussion, and also in relation to the comparison with McQueen et al, whose comparison from 2008-2015 would be similarly affected. The tables are comprehensive and helpful, but as formatted in the proof, extremely difficult to read. I assume the publisher will be able to do something with that? but because of this it would be helpful in the narrative to include a few more examples of how guidelines differ in other domains. In terms of differences in choice of particular medications, it is important to note that these may be governed local contracting, availability, and/or cost issues rather than therapeutic considerations and this should also be noted. As in the abstract, I feel this paper has reached some useful conclusions and these should be separated into a 'Conclusions' paragraph. Overall I welcome publication of this important study

VERSION 1 – AUTHOR RESPONSE

Point 1 – Please revise the ‘Strengths and limitations’ section of your manuscript (after the abstract). This section should contain up to five short bullet points, no longer than one sentence each, that relate specifically to the methods. The results of the study should not be summarised here.
Reply: Thank you for this comment. We revised the ‘Strengths and limitations” section and maintained only points related to the methods.
REVIEWER: 1 DR. A BAYES, UNSW
This paper seeks to examine CPGs for unipolar depression comparing and contrasting recommendations for patients with an inadequate response to first line treatment, using the AGREE-II tool. Overall while the paper was methodologically sound, the results found differences but were not particularly enlightening.
Reply: Thank you for the observation, at this time we tried to select and highlight the agreements and the most import differences among the CPGs to enrich the contribution of the manuscript.

Point 1 – Please justify why Recommendations for patients who did not respond to the first-line pharmacological treatment grouped according to patient reevaluation, dosing adjustment, switching, combination, and medication augmentation?
Reply: Thank you for this comment. In fact, initially, we have grouped the recommendations based on the most frequent strategies and also those that were regarded as the most relevant, which had been described in our comparison of the guidelines. However, after considering this major revised version of this work we have abandoned this procedure. Therefore, that approach has been modified. We included a description about this topic in the Materials and Methods section, as shown below:
Materials and Methods: “[…] The recommendations were grouped by the following main topics: terminology for responsiveness and recommended management strategies. The terminologies and sequences of the therapeutic strategies were compared between the CPGs and the strategies and terminologies that the CPGs had in common were synthesized in a third table.

Point 2 – Regarding method, it should specify if only unipolar depression is being examined (and bipolar depression is excluded) as there would be differences in recommendations between the two.

Reply: Thank you for this comment. We included only strategies recommended for unipolar depression. A sentence specifying this restriction has been added to the inclusion criteria in the Materials and Methods section. as shown below:
Materials and Methods: “[…]. CPGs for the treatment of bipolar depression only were also excluded.

Point 3 – The references in text seem to jump from 28 to 67. I could not see the references in superscript between these numbers.
Reply: Thank you for this comment. The references numbering system has been corrected. References that were only in the Tables are now also in the text.

Point 4 - The paper would benefit from discussion of limitations of guidelines themselves, that they are supposed to be 'evidence-based' but if the included evidence is of poor questionable quality, then the guidelines will reflect this.
Reply: Thank you for this comment. We included a sentence about this in our Discussion section, as follows:
Discussion: “[…]. “Another limitation to be considered is the questionable quality of evidence of primary efficacy studies for various therapeutic approaches, thus, weakness and disagreement among CPGs may at least in part reflect that condition.”

REVIEWER: 2 DR. LAUREN GERLACH, UNIVERSITY OF MICHIGAN
In this systematic review of clinical practice guidelines for patients who fail first line antidepressant treatment for depression, the authors highlight the significant variation and lack of consensus of current guidelines. Given the large impact of depression on quality of life for patients, costs of delivering care, and high prevalence of disease, as the authors highlight, more consensus is needed. Recommendations for improving the manuscript are below.
Reply: Thank you for the positive evaluation of our manuscript.

Major Concerns:
Point 1 – My biggest recommendation would be to reformat the Discussion section to focus more on what are the most common recommended treatment approaches and components of treatment based on the available current clinical practice guidelines. Even given the large variability present, discussing in more detail the range of treatment recommendations I think would be a helpful take away for readers and help make it more clinically relevant.
Reply: Thank you for this suggestion. We agree to the points you highlighted. The discussion section has been reviewed extensively to emphasize the agreement among the treatment approaches recommended by the included guidelines to make it more clinically relevant we also tried to discuss the convergencies and divergencies in more detail.

Point 2 – Additionally, any revised Table or Figure that could better summarize this material would be helpful for the reader. As is, the Tables presented are so large that they become difficult to read across multiple pages
Reply: Thank you for this comment, we also agree with them. The tables have been completely reframed to summarize our results more didactically.

Minor Comments:
Abstract:
Point 3 – Page 2, line 7: Please adjust wording of the Objective sentence (“depression differ for adults”). Similarly, the Conclusions statement could be improved.
Reply: We made the requested adjustments in both sections, as follows:
Objective: To assess similarities and differences in the recommended sequence of strategies among the most relevant clinical practice guidelines (CPGs) for the treatment of depression in adults with inadequate response to first-line treatment.

Conclusion: Relevant CPGs present shortcomings in recommendations for non-responders to first line antidepressant treatment including absence and divergencies in definition of inadequate response and sequence of recommended strategies. Overall, most relevant CPGs recommend reassessing the diagnosis, evaluate comorbidities and adherence to treatment, increase dosage of antidepressants and psychotherapy as first steps.

Point 4 – Page 3, line 6: Which two guidelines does this refer to?
Reply: Thank you for this comment. This information on which guidelines this sentence refers to has been added as follows:
Strengths and limitations of this study: The inclusion of three CPGs often used in clinical practice (from The Canadian Network for Mood and Anxiety Treatments – CANMAT; from the American Psychiatric Association – APA-Psychiatry; and from the US Department of Veterans Affairs (VA), US Department of Defense (DOD), enabled a broader discussion of clinical questions mentioned in the CPGs.
Introduction:
Point 5 – Page 4, line 22: For the authors statement: “Along with an increasingly aging population”, I would challenge the authors that depression prevalence among community dwelling (non-institutionalized) older adults (65+ years) is actually lower than that of general adults. Additionally, studies out of the U.S. suggest that older adults are also less likely than general adults to experience depression, anxiety, and insomnia following the start of the pandemic. Your point that COVID has been associated with increased mental health symptoms globally and therefore highlights the need for treatment services is well taken. However, I would not limit this discussion to just older adults.
Reply: Thank you for this comment. We reviewed the literature, and we agree with you. Therefore, we removed the statement about the aging population.

Methods:
Point 6 – Page 5, line 55: It would be helpful for readers not familiar to clearly specify what primary and secondary care for depression is. Does this include visits to outpatient psychiatrists or just primary care physicians?
Reply: Thank you for this comment. We included guidelines for outpatient settings, for either primary care physicians and psychiatrists. We reviewed the text in order to make this clearer for the readers. It reads:
Materials and Methods “[…] CPGs that can be used by health professionals for the pharmacological treatment of adults with depression in outpatient settings.”

REVIEWER: 3 PROF. PATRICK BALL, UNIVERSITY OF WOLVERHAMPTON
COMMENTS TO THE AUTHOR:
Thank you for the opportunity to review this interesting article. This review was undertaken in a competent and rigorous manner and highlights interesting limitations in the guidelines currently available and publication would be helpful.
Reply: Thank you for your very relevant comments and suggestions.

Areas where I would suggest minor revision are:
Point 1 – It is noted that there was high heterogeneity in some AGREE domains. Since many guidelines take up to five years from inception to publication, and DSM iV was replaced by DSM V in 2013, could some of this be due to variations in definitions applied? Perhaps this could be mentioned in the discussion, and also in relation to the comparison with McQueen et al, whose comparison from 2008-2015 would be similarly affected.
Reply: Thank you for this comment. We have based the definition of unipolar depression according to what had been established by the included guideline. During the period investigated there had been changes in the definition of depression and this has led to different level of prevalence of the diagnosis of depression. Although, when we applied the evaluation of the quality of guidelines using

AGREE II, this information is not considered among the domains of this tool. A change in the diagnostic criteria may affect the prevalence, even though it is unlikely to interfere on the evaluation of the AGREE II domains. We included in the discussion a brief paragraph in relation to this issue:
Discussion: Among the shortcoming of CPGs, this review reveals a high heterogeneity in quality of the rigour of development (domain 3 of AGREE). A difficult to explain finding. The Diagnostic and Statistical Manual of Mental Disorders (DSM) V replaced DSM IV in 2013, and the diagnostic criteria for depressive disorder have been updated. Such change could impact on case identification and estimative of depression prevalence. However, diagnostic criteria are not covered by AGREE II checklist and differences in quality among CPGs might have not been influenced by that change in DSM version. CPGs were from distinct years, and the APA-Psychiatry, published in 2010, the oldest included CPG, received the worst score on quality of rigour in development. It is possible that for the APA-Psychiatry and other CPGs the absence of a more recently updated version could have contributed to their low appraisal by AGREE-II

Point 2 – The tables are comprehensive and helpful, but as formatted in the proof, extremely difficult to read. I assume the publisher will be able to do something with that? but because of this it would be helpful in the narrative to include a few more examples of how guidelines differ in other domains.
Reply: Thank you for this comment. We reframed all the tables in order to facilitate the comprehension for readers.

Point 3 – In terms of differences in choice of particular medications, it is important to note that these may be governed local contracting, availability, and/or cost issues rather than therapeutic considerations and this should also be noted.
Reply: We agree with your assessment about this issue. Besides the clinical evidence, managers and decision-makers should also consider practical aspects such as the drug availability, the costs of the drugs, social and ethical values. Of note, recent methodologies for the development of guideline recommendations, such as the GRADE's Evidence-to-Decision (EtD) framework, provide a structured and transparent approach to use when formulating recommendations. We included a brief paragraph about this point in the discussion section of the paper:
"[…] Discussion: "It should be considered that discrepancies of choices of particular strategies or medications found in our review may be governed by local contracting, availability or cost issues besides evidence-to-decision (EtD) frameworks as it is recommended."

Point 4 – As in the abstract, I feel this paper has reached some useful conclusions and these should be separated into a 'Conclusions' paragraph.
Reply: Thank you, we made the recommended alterations.

Overall, I welcome publication of this important study.
Reply: Thank you very Much for this positive feedback!

Overall, we believe that in our revised manuscript, we have now carefully addressed all comments and concerns listed by the editors and expert Reviewers. The authors confirm that this manuscript has not been submitted to, and is not currently under review with, another journal.

We therefore shall be grateful if you would consider our revised manuscript for publication in your Journal.

VERSION 2 – REVIEW

REVIEWER	Bayes, A UNSW, Black Dog Institute
REVIEW RETURNED	22-Feb-2022

GENERAL COMMENTS	The authors have done a thorough job in addressing my queries. On reading the manuscript I note some grammatical/spelling issues have been added which should be rectified.

REVIEWER	Gerlach, Lauren University of Michigan
REVIEW RETURNED	23-Feb-2022

GENERAL COMMENTS	Thank you for your responsiveness to the Editor and Reviewer comments

REVIEWER	Ball, Patrick University of Wolverhampton, School of Pharmacy
REVIEW RETURNED	16-Feb-2022

GENERAL COMMENTS	Thank you the suggested minor revisions have been expertly addressed